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Internal presentation

- Vocab-size = 5
- Tokens = ["the", "cat", "is a", "dog", "ran"]

Token ID

- token_id("the") = 0
- token_id("cat") = 1
- token_id("ran") = 4

+++ token_embedding

let's say, we have hidden-size = 3

so, the token_embedding is a matrix of shape [vocab-size, hidden-size]

$$\Rightarrow [5, 3]$$

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		w0	w1	w2
"the"	(id 0)	0.1	0.2	-0.3
"cat"	(id 1)	0.9	0.4	0.7
"sat"	(id 2)	-0.2	0.5	0.1
"dog"	(id 3)	0.8	0.3	0.6
"ran"	(id 4)	-0.5	-0.1	0.4

If we want to look up

opt 1

$$\text{embedding("cat")} = \text{token-embedding}[1]$$

$$= [0.9, 0.4, 0.7]$$

[3]
shape

opt 2

using one-hot vector "cat"

$$\text{one-hot}[1] = [0.1, 0.0, 0.0, 0.0, 0.0]$$

Note: Its shape is [vocab-size] = [5]

Then:

$$\begin{aligned} \text{embedding} &= \text{one-hot}[1] @ \text{token-embedding} \\ &= [0.1, 0.0, 0.0, 0.0] @ [5, 3 \text{ matrix}] \\ &= [0.9, 0.4, 0.7] \quad [3] \end{aligned}$$

$$* [1, 5] @ [5, 3] = [1, 3]$$

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lm_head

hidden_size, [3]

we have let say. $h = [0.85, 0.35, 0.65]$

$$\star \text{ lm_head} = \text{token_embedding} = [\text{vocab_size}, h] \\ = [5, 3]$$

Then we can apply / trying the weight

- Then, compute the logit

$$\boxed{\text{logit} = h @ \text{token_embedding}^T} \\ \boxed{([1, 3] @ [3, 5] = [1, 5])}$$

computation

$$\bullet \text{ logit}[0] = h \cdot [0.1, 0.2, 0.3] = 0.005$$

$$\bullet \text{ logit}[1] = h \cdot [0.9, 0.4, 0.7] = 1.06$$

$$\bullet \text{ logit}[2] = h \cdot [-0.2, 0.5, 0.1] = 0.065$$

$$\bullet \text{ logit}[3] = h \cdot [0.8, 0.3, 0.6] = 1.175$$

$$\bullet \text{ logit}[4] = h \cdot [-0.5, -0.1, 0.4] = -0.16$$

$$\Rightarrow \text{logits} = [0.005, 1.06, 0.065, 1.175, -0.16]$$

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As a side note we can notice that $\text{count}[3]$ for the word "dog" is highest, simply because it happens to be a vector closer to "dog"'s embedding $[0.9, 0.3, 0.6]$

dot product = measure similarity